National University of Computer and Emerging Sciences
Karachi Campus

Database Systems (CS2005)

Date: Wednesday, 5th Nov 2025

Time: 08:45 to 09:45 am

Course Instructor(s): Dr. Zulfiqar Ali Memon, Dr. Farrukh Salim, Ms. Atiya Jokhio, Ms. Javeria Farooq, Mr. Basit Ali, Ms. Hajira Ahmed, Mr. Ali Naseer, Mr. Huzaifa, Mr. Umair

Mid 2 Exam

Total Time (Hrs):	**1**
Total Marks:	**15**
Total Questions:	**3**

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Attempt all the questions.

CLO # 3: Demonstrate an understanding of normalization theory to normalize the database and formulate, using SQL & relational algebra, solutions to a broad range of query & data problems.

Q1: [marks: 5] [Estimated Time: 20 minutes]

A hospital maintains the following relation to store information about patient visits:

Patient_Visit(VisitID, VisitDate, PatientID, PatientName, PatientAge, DoctorID, DoctorName, Department, RoomNo, Fee)

Scenario:

- Each **visit** has one unique VisitID and is handled by one **doctor**.
- A **patient** can visit many doctors, and a doctor can see many patients.
- Each **doctor** works in one **department** and one **room** and charges a fixed **fee** per visit.
- Patient's name and age depend on their **PatientID**.

1. Based on the scenario, **identify all possible functional dependencies (FDs)**. (2 marks)
2. Determine the **candidate key(s)** for the relation. (1 mark)
3. State the **highest normal form** the relation currently satisfies and **decompose** the relation into relations that satisfy **3NF**. (2 marks)

(1) Functional dependencies (from scenario)

- VisitID → VisitDate, PatientID, DoctorID
  (each visit has one date, one patient, one doctor)
- PatientID → PatientName, PatientAge
  (patient attributes determined by patient id)
- DoctorID → DoctorName, Department, RoomNo, Fee
  (each doctor has fixed name, department, room and fee)

By transitivity: VisitID → DoctorID → Fee etc., but primitive FDs listed above are the main ones.

(2) Candidate key(s)

- **Candidate key:** VisitID — it uniquely identifies a visit and (via the FDs) determines all other attributes.

(3) Highest normal form of the given relation (brief justification)

- **1NF:** Yes — attributes are atomic.
- **2NF:** Yes — primary key is single attribute VisitID, so no partial dependencies.
- **3NF: No** — there are **transitive dependencies**: VisitID → DoctorID and DoctorID → Fee implies VisitID → Fee through a non-key attribute (DoctorID) → violates 3NF.

Conclusion: relation is **in 2NF but not in 3NF**.

Decomposition to 3NF (step & brief justification)

Decompose to remove transitive dependencies:

1. **Visits**(VisitID **PK**, VisitDate, PatientID, DoctorID)
   - FD: VisitID → VisitDate, PatientID, DoctorID
2. **Patients**(PatientID **PK**, PatientName, PatientAge)
   - FD: PatientID → PatientName, PatientAge
3. **Doctors**(DoctorID **PK**, DoctorName, Department, RoomNo, Fee)
   - FD: DoctorID → DoctorName, Department, RoomNo, Fee

CLO # 2: Analyze an information storage problem and derive an information model expressed in the form of an entity relation diagram and other optional analysis forms, such as a data dictionary.

Q2: [marks: 6] [Estimated Time: 25 minutes]

A soccer association wants to design a database system to manage information about teams, their players, the matches they play, and the referees who supervise those matches. Teams are uniquely identified by TeamID. For each team, store TeamName, HomeStadium, and City. A team may participate in many matches either as a Host Team or as a Guest Team. Players play matches and Each Player belongs to exactly one team. For every player, store PlayerID, PlayerName, DateOfBirth, StartYear, and ShirtNumber. Shirt numbers are unique within a team, but may repeat across different teams.Matches have a unique MatchID. Record MatchDate and FinalScore. The final score is stored as two numeric attributes (**HostGoals**, **GuestGoals**).
A match is played at the stadium of the host team. Each match involves: One Host Team, One Guest Team, Exactly three referees: one Main Referee and two Assistant Referees Players from both teams may participate in a match. For each player participating in a match, record: GoalsScored, YellowCard (Yes/No), RedCard (Yes/No). Only players belonging to either the Host or Guest team may participate in that match.
Referees are identified by RefereeID. For each referee, store RefereeName, DateOfBirth, and YearsOfExperience. A referee may officiate many matches and may serve in different roles (Main or Assistant) in different matches.

1. Identify the entities from the scenario. (1 mark)
2. Define the attributes for each entity based on the specific requirements. (1 mark)
3. Determine the relationships between entities. (1 mark)
4. Draw the complete Entity Relationship Diagram (ERD). Clearly mention the structural constraints i.e. cardinalities and participation. (3 marks)

Solution:
Part 1, 2 and 3
Entities, Attributes and relationships:

1. Team

- TeamID (PK)
- TeamName
- HomeStadium
- City

2. Player

- PlayerID (PK)
- PlayerName
- DateOfBirth
- StartYear
- ShirtNumber (unique within team)
- TeamID (FK)

3. Match

- MatchID (PK)
- MatchDate
- HostGoals
- GuestGoals
- HostTeamID (FK)
- GuestTeamID (FK)

4. Referee

- RefereeID (PK)
- RefereeName
- DateOfBirth
- YearsOfExperience

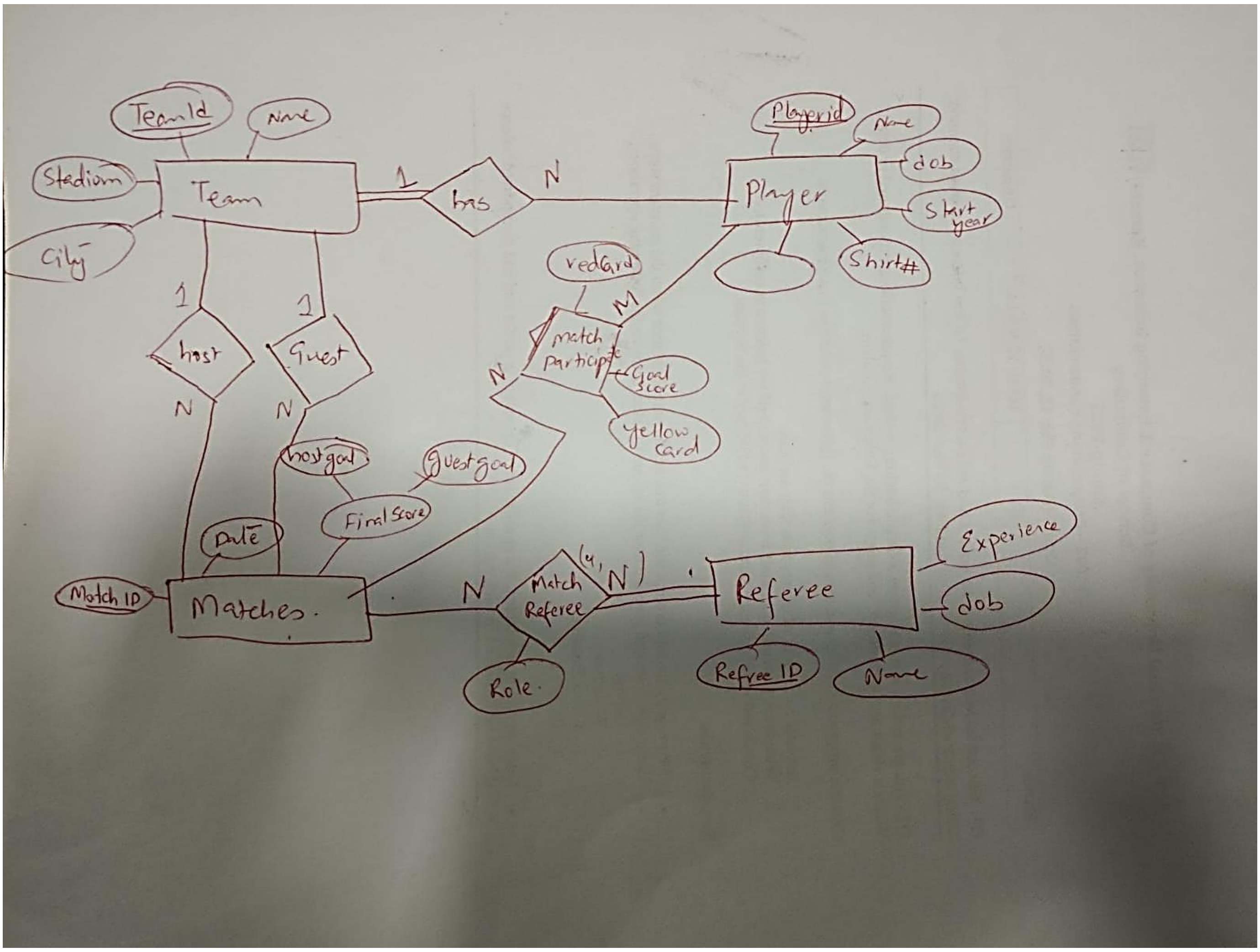
5. MatchParticipation

- MatchID (FK)
    - PlayerID (FK)
    - GoalsScored
    - YellowCard (Yes/No)
    - RedCard (Yes/No)

6. MatchReferee

- MatchID (FK)
- RefereeID (FK)
- Role (Main / Assistant)

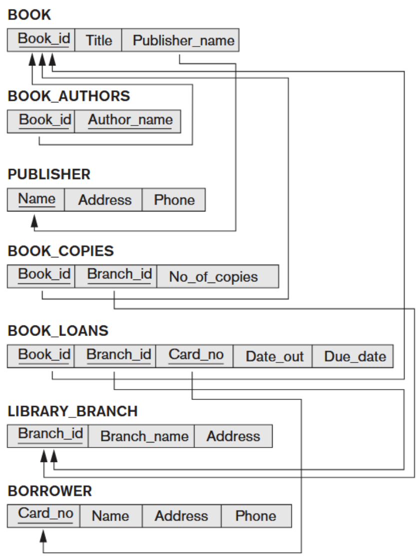
Relationships:
- Team–Match: One team can be Host or Guest in many matches.
- Player–Team: Each player belongs to one team.
- Player–Match: Many-to-many via MatchParticipation.
- Referee–Match: Many-to-many via MatchReferee with role attribute.



CLO # 3: Demonstrate an understanding of normalization theory to normalize the database and formulate, using SQL & relational algebra, solutions to a broad range of query & data problems.

Q3: [marks: 4] [Estimated Time: 15 minutes]

Consider the LIBRARY relational database schema given, which is used to keep track of books, borrowers, and book loans. Write down relational algebra expressions for the following queries:



1. How many copies of the book titled The Lost Tribe are owned by the library branch whose name is 'Sharpstown'?

$$\mathcal{F}_{\text{SUM(No\_of\_copies)}}\left(\sigma_{\text{Title='The Lost Tribe'} \wedge \text{Branch\_name='Sharpstown'}}\left(\text{BOOK} \bowtie \text{BOOK\_COPIES} \bowtie \text{LIBRARY\_BRANCH}\right)\right)$$

2. Retrieve the names of all borrowers who do not have any books checked out.

$$\pi_{\text{Name}}(\text{BORROWER}) - \pi_{\text{Name}}(\text{BORROWER} \bowtie \text{BOOK\_LOANS})$$

3. For each book that is loaned out from the Sharpstown branch and whose Due_date is today, retrieve the book title, the borrower's name, and the borrower's address.

$$\pi_{\text{Title, Name, Address}}\left(\sigma_{\text{Branch\_name='Sharpstown'} \wedge \text{Due\_date=TODAY}}\left(\text{BOOK} \bowtie \text{BOOK\_LOANS} \bowtie \text{BORROWER} \bowtie \text{LIBRARY\_BRANCH}\right)\right)$$

4. For each library branch, retrieve the branch name and the total number of books loaned out from that branch

$$_{\text{Branch\_name}}\mathcal{F}_{\text{COUNT(Book\_id)}}\left(\text{BOOK\_LOANS} \bowtie \text{LIBRARY\_BRANCH}\right)$$